Project report - Remotely controlled airplane

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1 Project Description

As part of the project for Module B, I built a remote-controlled model airplane. The airplane is made from extruded polystyrene, plastic, adhesive foil, and wood, which are easily accessible and relatively inexpensive materials. Generic components, including an Arduino Nano, a generic brushless motor, a 30A speed controller, SG90 servomotors, and a 3S 1300mAh LiPo battery, are used to power the airplane and control its functions.

The designed model is relatively small in size, with a wingspan of 80 cm, a length of 70 cm, and a weight of approximately 500 g. It is shown in figure 1

2 Materials and Components

In this section, we will discuss the materials and components used in the construction of the remote-controlled model airplane.

2.1 Structural/Mechanical Materials

The following materials were used to make the structural parts of the airplane and various mechanical components:

- **6mm Depron (extruded polystyrene):** This is a lightweight, durable material used for floor insulation. Due to its high strength-to-weight ratio and easy workability, it is ideal for making structural parts such as the fuselage and wings of the airplane.
- **Adhesive foil for notebooks:** This material was used to cover and protect the external surfaces of the airplane, providing tensile strength.
- **8060 propeller:** An 8-inch propeller for powering the airplane.
- **3mm plywood:** This was used to make sturdier parts of the airplane, such as the motor mount, wing support, wheels, and control wire attachment points.
- **Steel wire (1mm and 1.8mm thick):** These wires were used to make control rods and attach certain electronics (1mm) and to make the landing gear (1.8mm).
- **Paper straws:** These were used as a lightweight and durable solution for guiding control wires through the fuselage of the airplane.
- **Hot glue:** This was used to permanently attach various parts of the airplane.

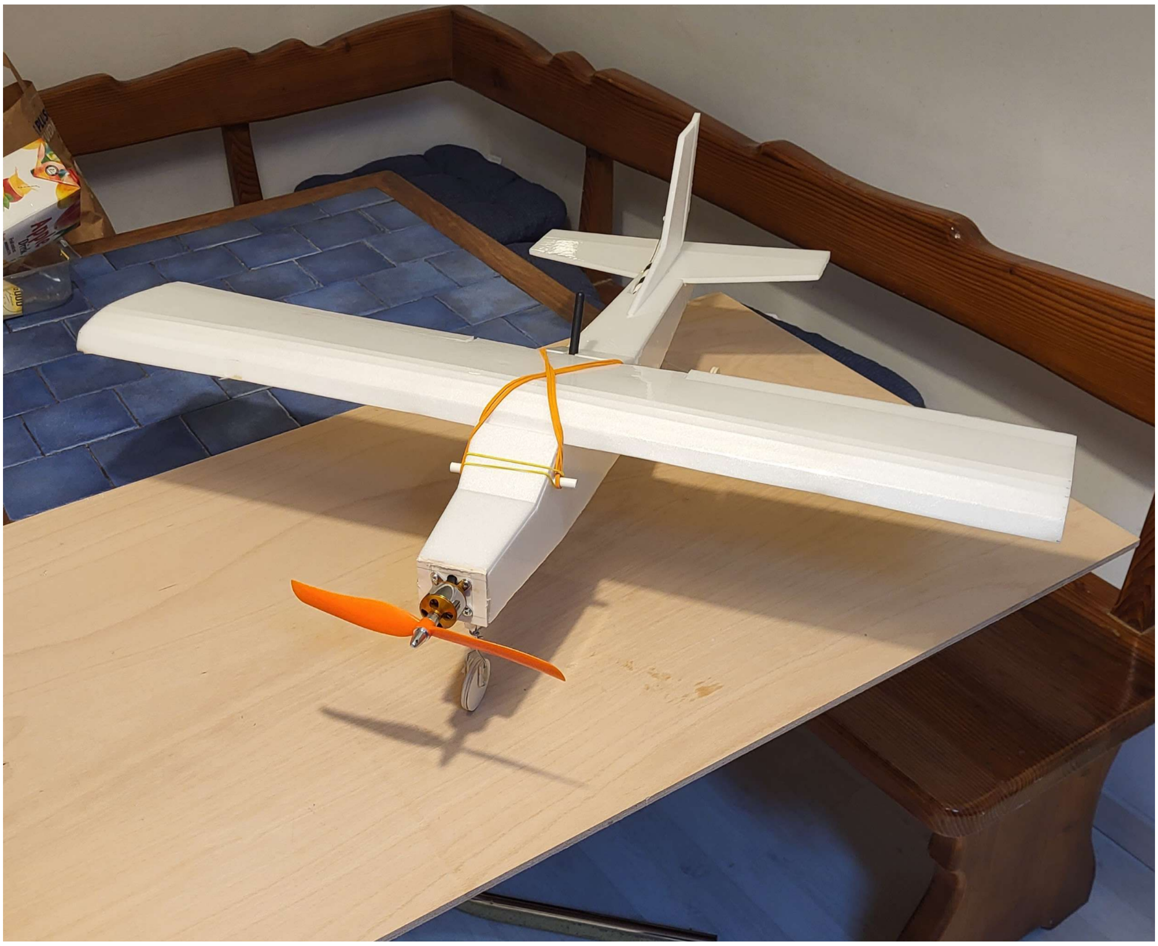

Figure 1: Model airplane built as part of the project

- **Elastic bands:** These were used to attach the wings to the fuselage of the airplane, allowing for easy assembly and disassembly.
- **M3x16 screws, nuts, and washers:** These were used to attach the motor to the airplane.
- **7mm PVC pipe:** A lightweight and robust solution for wing attachment points.

2.2 Electronic Components

The following components were used for the electrical operation of the model airplane:

- **Arduino Nano:** This microcontroller was used as the brains of the entire system.
- **A2212 1400KV brushless motor:** This was used as the airplane's propulsion.
- **30A ESC (electronic speed controller):** This component is used to regulate the rotation speed of the brushless motor.

- **SG90 servomotors:** These were used to control the airplane's control surfaces, such as ailerons and rudder.
- **LiPo 3S 1300mAh battery:** This is the main power source for all the airplane's electrical components, including the motor, microcontroller, and servomotors.
- **LM2596 voltage regulator:** A switching regulator, it brings the battery voltage down to a level suitable for the operation of other components.
- **NRF24L01+PA+LNA 2.4GHZ transceiver:** This transceiver was used for wireless communication between the remote controller and the airplane.
- **Copper wire (cross-section 0.34mm^2, 2.5mm^2):** The 0.34mm^2 wire was used to transmit signals and power all components except the main motor, for which the 2.5mm^2 wire was used.
- **XT60, T, and 2.54mm connectors:** These connectors were used to connect various electrical components in the system, allowing for easy assembly and component replacement.
- **Joystick:** For the radio transmitter, the airplane is controlled via two joysticks.

3 Connection List

The power supply for the airplane comes from a single LiPo 3S (11.1V) battery. The positive pole (+) of the battery is connected to the +IN input of two LM2596 voltage regulators, which generate 5V and 7V, and to the + input on the ESC (Electronic Speed Controller). The negative pole (-) of the battery is also connected to the -IN input of the same regulators and to the - input on the ESC.

The outputs of the voltage regulators power various parts of the airplane. The 5V output and common GND are connected to the VCC and GND of all servomotors and to the VCC and GND of the ESC. The 7V output and GND go to the VIN and GND of the Arduino Nano controller.

The three-phase output of the ESC is connected to the motor and provides clockwise rotation of the motor.

The ESC (which controls the drive motor) is connected to the digital pin D2 on the controller. The servo for the rudder is connected to D3, the servo for the elevator is connected to D4, and both servos for the ailerons are connected to D5.

Finally, we have connections for the 2.4 GHz NRF24L01 transceiver. These connections are as follows:

- The VCC of the transceiver is connected to the 3.3V pin on the Arduino Nano.
- The GND of the transceiver is connected to the GND on the Arduino Nano.
- The MOSI of the transceiver is connected to D11 on the Arduino Nano.
- The MISO of the transceiver is connected to D12 on the Arduino Nano.
- The SCK of the transceiver is connected to D13 on the Arduino Nano.
- The CSN of the transceiver is connected to D10 on the Arduino Nano.
- The CE of the transceiver is connected to D9 on the Arduino Nano.

All of this ensures that our airplane is properly powered, can receive and send data via the transceiver, and can correctly control the motor and servomotors.

4 Program

4.1 Transceiver Configuration

The configuration of the 2.4 GHz NRF24L01 transceiver requires communication via SPI (Serial Peripheral Interface) and the correct setting of various parameters. The basic initialization includes the following steps, which must be performed in the order listed:

1. Initialization of SPI, which includes setting the MOSI, SCK, and CSN pins to output, the MISO pin to input, turning on SPI in master mode, and setting the frequency to CLK/16 = 1 MHz. This is the highest frequency that the transceiver supports.
2. Setting the CE and CSN pins to output.
3. Setting the CE pin to low state.
4. Setting the CSN pin to high state.
5. Waiting 5 ms for the transceiver to start.
6. Setting the following parameters in the transceiver via SPI: power off, no CRC, display interrupts on the IRQ pin (not in use), turn off auto acknowledgement, turn off all data pipes, set the address width to 5 bytes, turn off retransmission, set the channel to 0, set the power attenuation to 0 dB, set the data rate to 2 Mbps.
7. Setting the CE pin to high state.
8. Setting the CSN pin to low state.

For complete configuration on the airplane side, which we configure as a receiver, the following additional steps must be performed:

1. Setting the CE pin to low state.
2. Setting the CSN pin to high state.
3. Selecting the channel (in our case 10).
4. Turning on the first data pipe.
5. Setting the receiving address (in our case `0xEEDDCCBBAA`).
6. Setting the packet width to 32 bytes, which is the maximum value.
7. Turning on the device.
8. Waiting 2 ms.
9. Setting the CE pin to high state.
10. Setting the CSN pin to low state.

With this, our transceiver is fully configured for data reception.

4.2 Data Reception

For data reception, the system performs polling to see if a packet has arrived. When the system receives a data packet, it is stored in a temporary table for further processing. The data is read via SPI using a read command. A data packet consists of 32 uint8_t values, out of which the first 4 are joystick readings for thrust, yaw, pitch and roll, respectively. The remaining 28 bytes are unused in the current implementation.

4.3 Servo and Motor Control

The servos and motor of the airplane are controlled via pulses with a frequency of 50 Hz. The length of these pulses determines the rotation of the servos and the speed of the motor. The length of the pulses for the servos ranges between 0.5 ms and 2.5 ms (0° to 180°), and for the ESC between 1 ms and 2 ms (0% - 100%).

The pulses are generated as follows: a 20 ms period is divided into five subperiods, corresponding to pins D2, D3, D4, and D5 and all off. The state changes when the time set for the previous task has elapsed. The temporary table with the received data is used at the end of the cycle to recalculate the number of clock cycles corresponding to the received signal. For the servos, the joystick value is mapped with a simple linear mapping, for the ESC all values below 140 are considered 0, which corresponds to approximately 200mV safety margin before the motor turns on. The signal sequence is shown in figure 2.

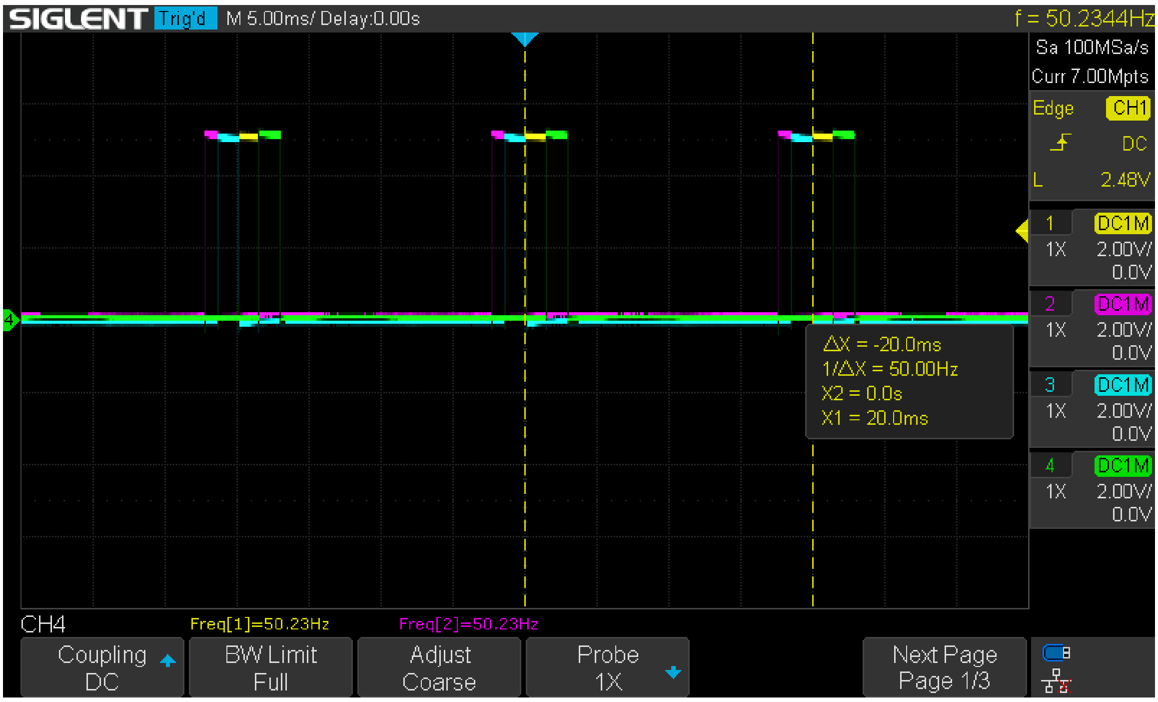

Figure 2: Control signals for servo motors.

5 Radio Transmitter

The airplane is controlled with a specially made radio transmitter. The radio transmitter uses the same material and has a fairly similar connection. The important difference is that everything is powered from the controller. It also uses an Arduino Nano, to which the transceiver is connected in exactly the same way. In addition, it uses two XY joysticks. These are connected as follows:

- The 5V pin on the joysticks is connected to the 5V pin on the Arduino Nano,
- The GND pin on the joysticks is connected to the GND pin on the Arduino Nano,
- The VRy pin of the left joystick is connected to the A0 pin on the Arduino Nano,
- The VRx pin of the left joystick is connected to the A1 pin on the Arduino Nano,
- The VRy pin of the right joystick is connected to the A2 pin on the Arduino Nano,
- The VRx pin of the right joystick is connected to the A3 pin on the Arduino Nano.

The transmitter's transceiver is configured slightly differently:
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1. Basic configuration
2. Setting the CE pin to low state.
3. Setting the CSN pin to high state.
4. Selecting the channel (in our case 10).
5. Setting the transmission address (in our case `0xEEDDCCBBAA)`).
6. Turning on the device.
7. Waiting 2 ms.
8. Setting the CE pin to high state.
9. Setting the CSN pin to low state.

The controller reads the joystick values with 8-bit resolution. All 4 values are written to a table and sent via the transmitter. The transmission speed is approximately 20Hz (more is not sensible due to the limitations of the servos, the theoretical maximum is 50Hz but they start to jerk, the solution is in better joysticks and servos).

6 Development Issues

The main issue was the configuration of the transceiver, as it is difficult to debug. Also, from the documentation, it was not entirely clear what the minimum working configuration was. The problem was alleviated by reading the status of the registers on the transceiver and analyzing existing libraries. Existing servo libraries were also not compatible with this implementation, so the servo control had to be written from scratch. Other issues were primarily hardware-related (destroyed components due to incorrect power supply, unstable operation, and controller resets during current pulses). These were resolved with improved power supply that completely isolates all power parts of the circuit from the signal ones.

7 Possible Improvements

The current model airplane is primarily a prototype and learning experience, but it is not suitable for any tasks. This is primarily due to the compact size, which does not allow much room for additional equipment. The next iteration should be at least 2x larger or further shrink the electronics (difficult to implement). The current transceiver has a range of about 1km. Alternatives are a more powerful transceiver or control via mobile networks. For a larger range, a real-time camera would also be necessary for the possibility of controlling outside the field of view.

8 Additional Documentation

The properties of the components are more accurately described in their data sheets. A more detailed explanation of the code is in the comments in the attached source code.